INTERNET OF THINGS

Optimizing Sink Position in a Wireless Sensor Network

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Problem Statement

In the parking lot there are 10 sensors that monitor the parking spaces. Each sensor has a fixed position (x,y) in the parking lot reported in Table.

Sensor	Coordinates (x,y)
1	(1, 2)
2	(10, 3)
3	(4, 8)
4	(15, 7)
5	(6, 1)
6	(9, 12)
7	(14, 4)
8	(3, 10)
9	(7, 7)
10	(12, 14)

Given Information

Each sensor transmits a status update every 10 minutes, consuming energy per transmission.

- ❖ E_b=5 mJ=5,000,000 nJ (Initial engergy per sensor)
- ❖ E_c=50 nJ/bit (circuitry energy for TX/RX).
- ❖ b=2000 bits (packet size).
- ❖ k=1 nJ/bit/m² (energy coefficient).

The energy consumption for transmission depends on the distance between the sensor and the sink.

Objective A

The objective is to find the system lifetime when the sink is placed at (20,20).

Computing the distance of Each Sensor to the Sink at (20,20)

$$d_i = \sqrt{(x_s - x_i)^2 + \left(y_s - y_i\right)^2}$$

$$d_{(20,20)} = \sqrt{(20 - x_i)^2 + (20 - y_i)^2}$$

Sensor 1- Coordinates(x,y) =(1,2)

$d_{(1,2)} = \sqrt{(20-1)^2 + (20-2)^2}$ = 26.19

Sensor 2- Coordinates(x,y) =(10,3)

$d_{(10,3)} = \sqrt{(20-10)^2 + (20-3)^2}$ = 19.72

Sensor 3- Coordinates(x,y) =(4,8)

$d_{(4,8)} = \sqrt{(20-4)^2 + (20-8)^2}$ = 20

Sensor 4- Coordinates(x,y) =(15,7)

$d_{(15,7)} = \sqrt{(20-15)^2 + (20-7)^2}$ = 13.93

Sensor 5- Coordinates(x,y) =(6,1)

$d_{(6,1)} = \sqrt{(20-6)^2 + (20-1)^2}$ = 23.6

Sensor 6- Coordinates(x,y) =(9,12)

$d_{(9,12)} = \sqrt{(20-9)^2 + (20-12)^2}$ = 13.6

Sensor 7- Coordinates(x,y) =(14,4)

$d_{(14,4)} = \sqrt{(20-14)^2 + (20-4)^2}$ = 17.09

Sensor 8- Coordinates(x,y) =(3,10)

$d_{(3,10)} = \sqrt{(20-3)^2 + (20-10)^2}$ = 19.72

Sensor 9- Coordinates(x,y) =(7,7)

$d_{(7,7)} = \sqrt{(20-7)^2 + (20-7)^2}$ =18.38

Sensor 10- Coordinates(x,y) =(12,14)

$d_{(12,14)} = \sqrt{(20-12)^2 + (20-14)^2}$ = 10

Formla for compute total energy per sensor

$Etotal = Ec * b + k\ di^2 * b$

Ec*b=(50×2000)=100,000 nJ

di^2 *b=(1×di^2×2000)=2000×di^2

$$Etotal = 100{,}000 + 2000 \times di^2$$

Formula for compute the sensor lifetime

Ti=$\frac{E_b}{E_{total}}$

$$Ti = \frac{5{,}000{,}000}{Etotal}$$

sensor	di^2	$Etotal$	Ti(Transmission cycle)
1	685	14,70,000	3.4
2	389	8,78,000	5.69
3	400	9,00,000	5.56
4	194	4,88,000	10.25
5	557	12,14,000	4.12
6	185	4,70,000	10.64
7	292	6,84,000	7.31
8	389	8,78,000	5.69
9	338	7,76,000	6.44
10	100	3,00,000	16.67

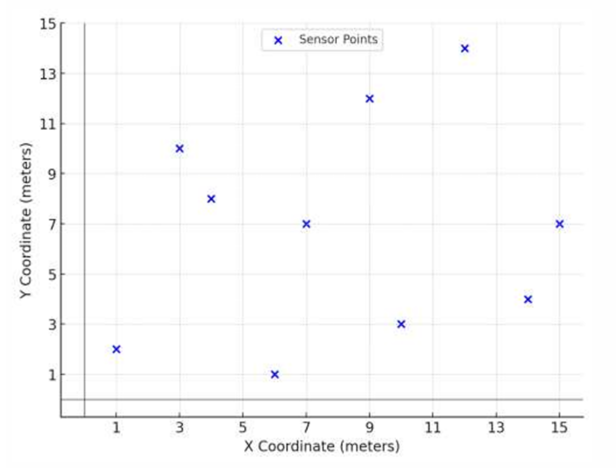
The sensor with the shortest lifetime determines the overall system lifetime.

From calculations, Sensor 1 at (1,2) has the shortest lifetime of 3.40 transmission cycles. Since each cycle occurs every 10 minutes, the total system lifetime before failure is:

System Lifetime=3.40×10 = 34 minutes.

Objective B

The goal is to determine the optimal sink position that maximizes system lifetime by minimizing the energy consumption of the worst-case sensor. The worst-case sensor is the one that consumes the most energy per transmission, leading to the shortest lifetime.



Optimized Sink Position

Applying numerical optimization, the best sink location was identified as:

(x_s, y_s)=(6.87,7.66)

This position was determined through:

- ❖ Distance recalculations between sensors and the sink.
- ❖ Energy consumption evaluation based on Objective A's calculations.
- ❖ Iterative refinement via the Nelder-Mead algorithm to minimize the worst-case sensor's energy depletion.

At the optimized sink position (6.87, 7.66)

- ❖ The worst-case sensor's energy consumption is minimized, allowing it to operate for a longer duration.
- ❖ System lifetime improves from 34 minutes (sink at 20,20) to approximately **214 minutes, a 6-fold increase.**
- ❖ Energy usage is more evenly balanced, reducing the likelihood of early sensor failures.

By repositioning the sink to **(6.87, 7.66),** we achieve:

- ❖ **Extended network longevity**, ensuring continuous and reliable operation.

- ❖ **Optimized energy distribution**, reducing strain on the most energy-intensive sensor.
- ❖ **Significant improvement in efficiency**, making the system more sustainable for long-term use.

Objective C

The objectve is to discuss trade-offs Between Fixed Sink Position versus Dynamically Moving Sink when designing a Wireless Sensor Network.'

Fixed Sink Placement

Advantages:

- ❖ Simplicity and Ease of Implementation: A fixed sink requires no mobility algorithms, making it easy to deploy and maintain.
- ❖ Predictable Communication Patterns: Since the sink's location remains constant, routing protocols and network configurations remain stable.
- ❖ Lower Energy Overhead: No additional energy is required for moving the sink, which is beneficial in resource-constrained environments.

Disadvantages:

- ❖ Unequal Energy Depletion: Sensors farther from the sink consume significantly more energy than closer sensors, leading to premature failure of distant nodes.
- ❖ Shorter System Lifetime: As some sensors deplete their energy faster, network coverage degrades over time.
- ❖ Less Adaptability: A fixed sink cannot adjust to dynamic environmental changes, sensor failures, or varying network conditions

Dynamic Sink Placement

Advantages:

- Energy Load Balancing: The sink can be repositioned to distribute the energy consumption more evenly across the network, reducing strain on distant sensors.
- Extended System Lifetime: By moving closer to energy-depleted sensors, a dynamic sink can significantly increase network longevity.
- Adaptability to Changing Conditions: The sink can adjust its position based on traffic patterns, sensor failures, or new node deployments, optimizing data collection efficiency.
- Disadvantages:
- Complex Implementation: Mobility algorithms and real-time decision-making are required, increasing system complexity.
- Additional Energy Costs: Moving the sink consumes energy, which must be accounted for in system design.
- Potential Communication Delays: Sink mobility can lead to temporary disconnections or increased routing complexity, affecting real-time data transmission.

Feature	Fixed Sink Placement	Dynamic Sink Placement
Ease of Implementation	High (Simple deployment)	Low (Requires mobility algorithms)
Energy Distribution	Unequal (Sensors far from sink consume more energy)	Balanced (Sink moves to optimize energy use)
System Lifetime	Shorter due to rapid depletion of distant sensors	Longer as energy load is distributed more evenly
Network Adaptability	Low (Not responsive to failures or changes)	High (Can adjust to environmental factors)
Energy Overhead	Low (No movement required)	High (Mobility incurs energy costs)

Conclusion

- The choice between fixed and dynamic sink placement depends on the specific network requirements:
- A fixed sink is ideal for simple, low-maintenance networks where energy efficiency is not a primary concern.
- A dynamic sink is best for maximizing system lifetime and balancing energy consumption across the network but requires additional complexity.
- A hybrid approach, where the sink moves periodically based on energy depletion patterns, could offer a balanced trade-off between simplicity and efficiency.
- Ultimately, strategic sink placement is critical for optimizing network performance, prolonging system lifetime, and ensuring sustainable energy consumption in wireless sensor networks.